



Blockchains, blockweaves, and the decentralised revolution

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Sam Williams, Archain CEO



My background

- CEO of the Archain project
 - A decentralised, permanent archive for the internet
 - <https://www.archain.org>
- Erlang/distribution specialisation
- PhD candidate
 - Developed HydrOS, a distributed operating system, written in Erlang
 - <http://hydros-project.org>



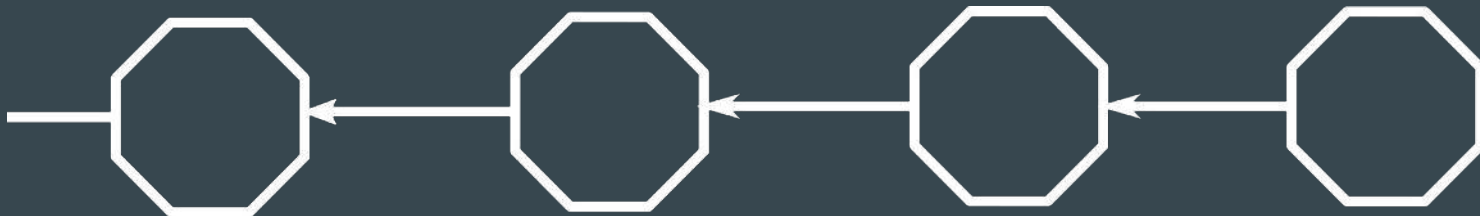
Presentation summary

- Blockchains & blockweaves
 - Their similarities & differences
- The Archain project - an open, immutable, decentralised archive
 - Potential effects of the Archain
 - Building Archain
 - Erlang's role
 - Testing the Archain network
- A new model for decentralised application design
 - Archain app competition

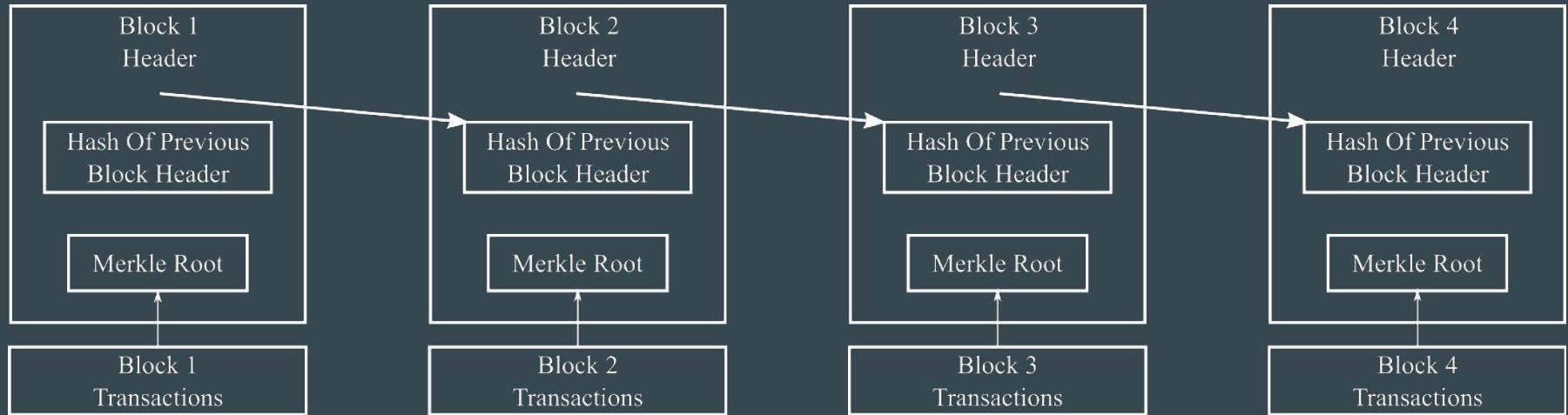


Blockchain Basics: What are blockchains?

- Hosted on a peer to peer network.
- A decentralised system for negotiating and storing transactions
- Records ('blocks') are comprised of transactions
- Blocks are linked and secured by cryptography
- Blocks are added by participating in a 'Proof of Work' system
 - 'Miners' generate random chunks of data and hash them with an 'open' block, looking for a result with a certain number of preceding zeros

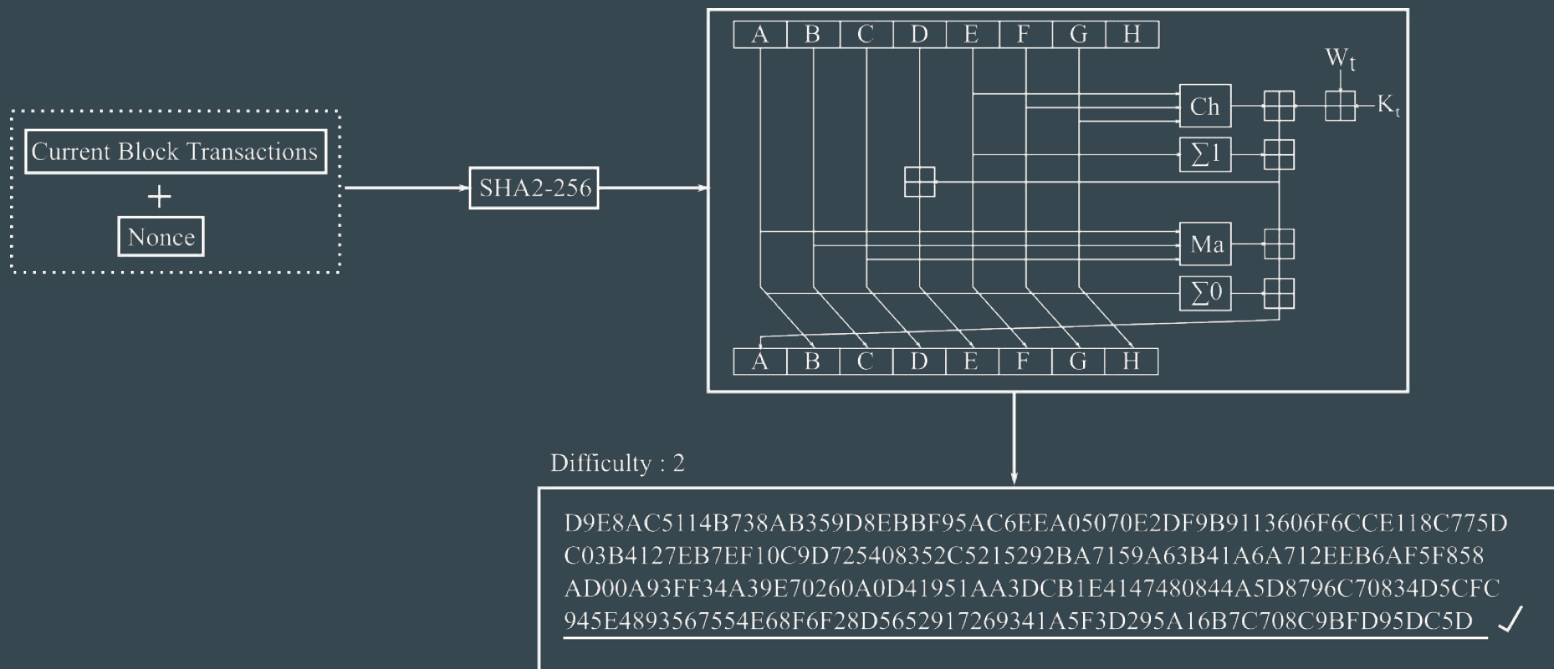


Blockchain Basics: What are blockchains?



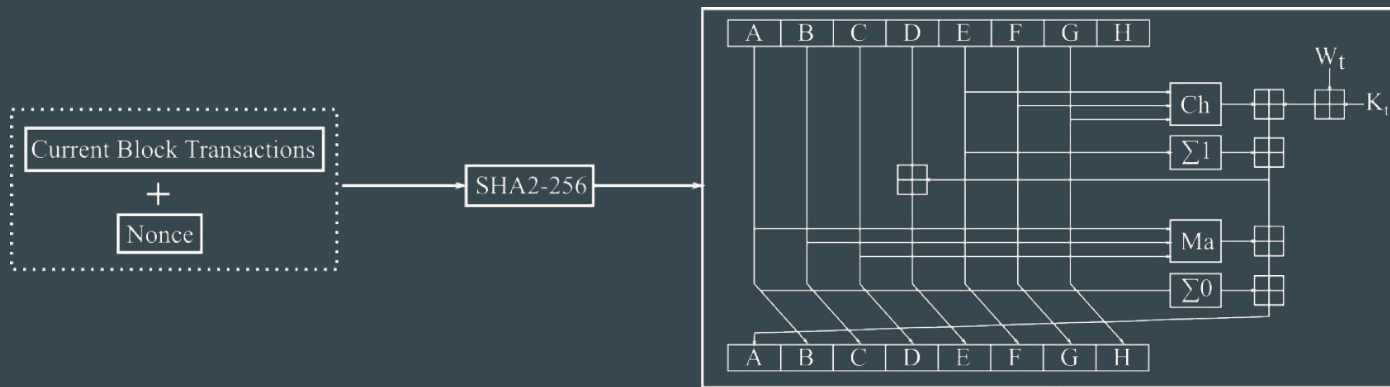


Blockchain Basics: Proof of Work (easy)





Blockchain Basics: Proof of Work (hard)



Difficulty : 4

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C03B4127EB7EF10C9D725408352C5215292BA7159A63B41A6A712EEB6AF5F858
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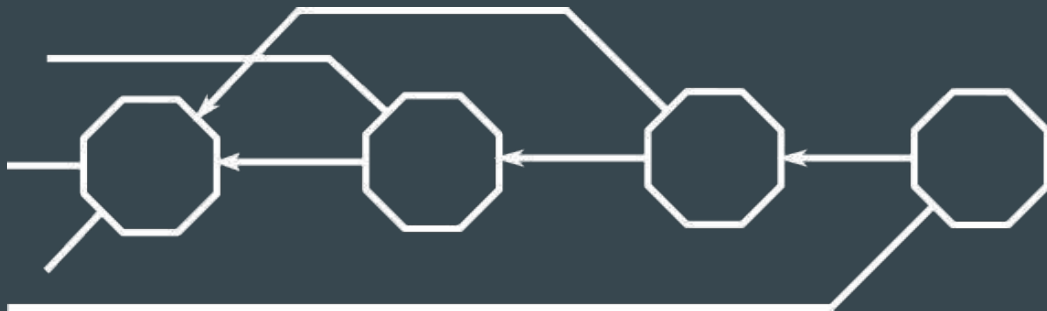
Blockchain Basics: Why are blockchains important?

- Transactions can be made without any kind of centralised authority
- Transactions are cryptographically verifiable at all times
 - Monetary system secured by cryptography, not trust
- Examples of use:
 - Bitcoin - for value transfer
 - Ethereum - for smart contracts



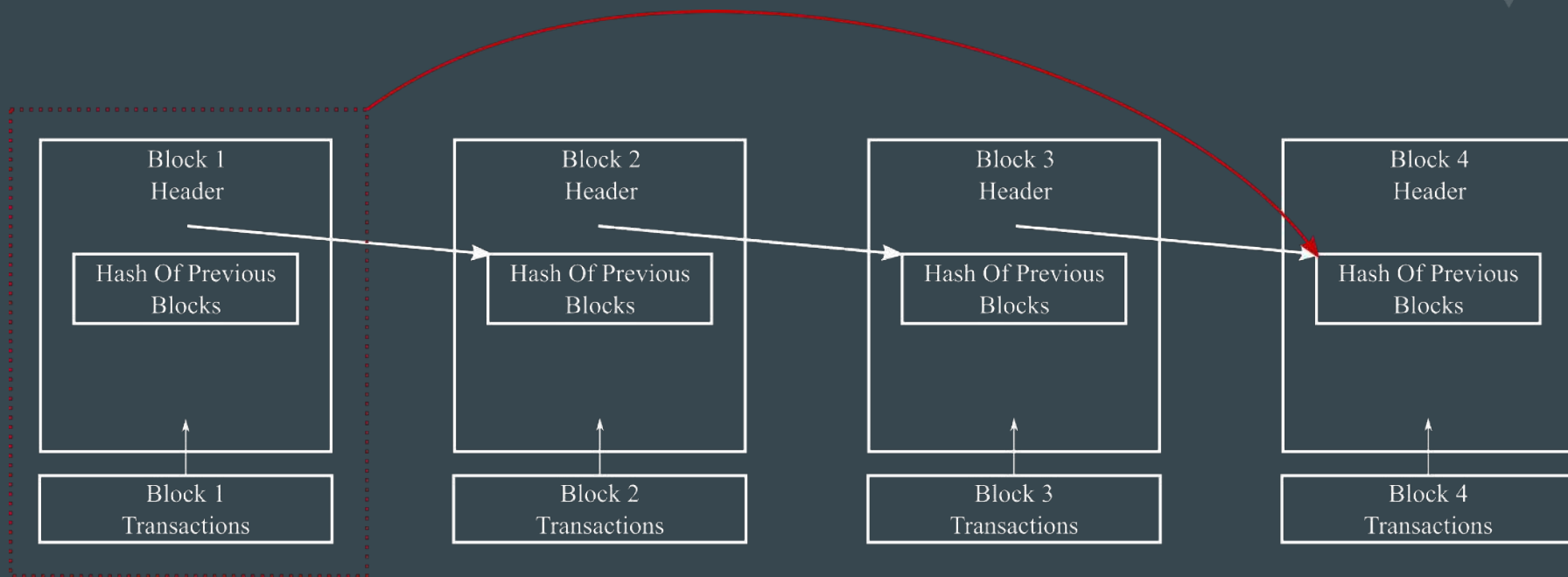
Blockweave Basics: What are blockweaves?

- Sharding
 - A blockweave is a blockchain that shards
- Proof of Access
 - Built on top of typical Proof of Work algorithms
 - Replaces most 'useless' work with **useful** storage
 - Maintains PoW security guarantees
- Distributed storage
 - Evenly-distributed & self-organising storage by miners

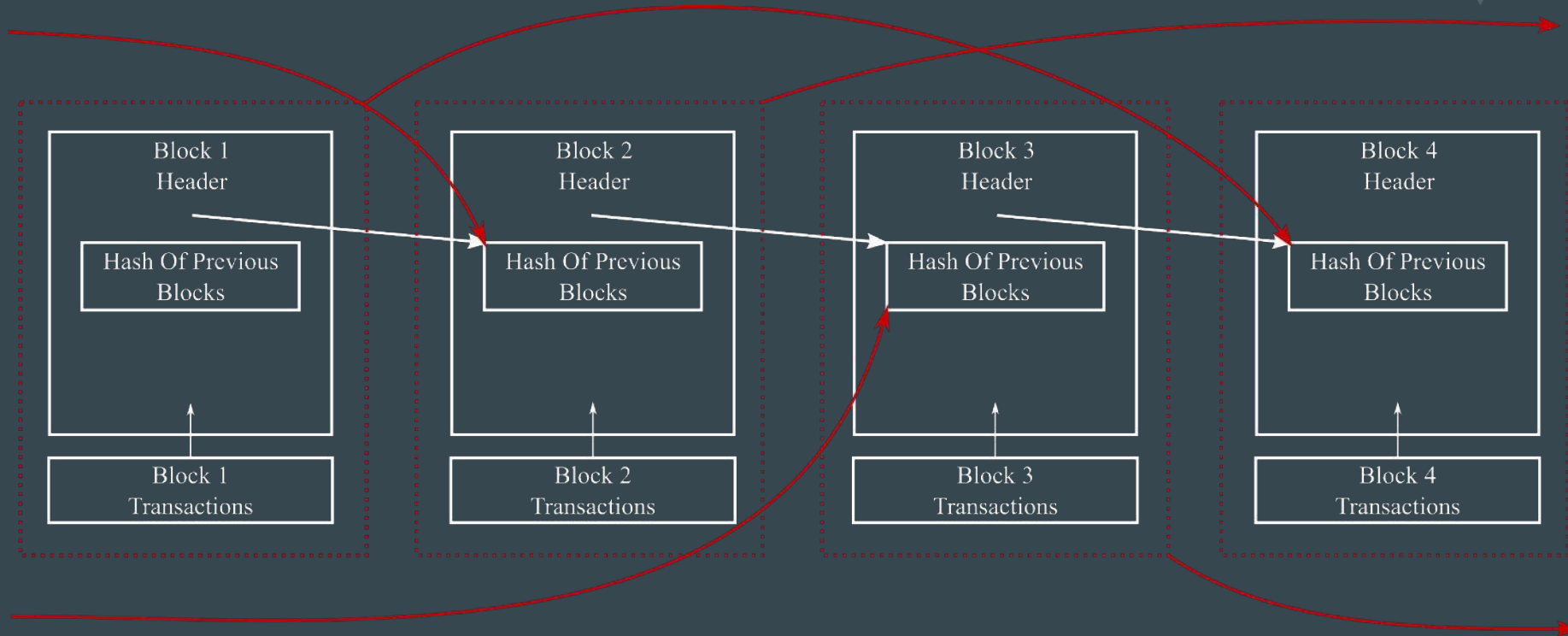




Blockweave Basics: What are blockweaves?



Blockweave Basics: What are blockweaves?





Blockweave Basics: Why are blockweaves better?

- Sharding
 - Store data directly on the blockweave
 - Allows for huge scaling potential
- Convenience for miners
 - Miners can instantly join the network and mine almost immediately
 - Download just a few blocks to start, instead of >200GB
- Consensus
 - Datastructure can be split among many machines on the network
 - Allowing for consensus and storage to be achieved for much larger pieces of information



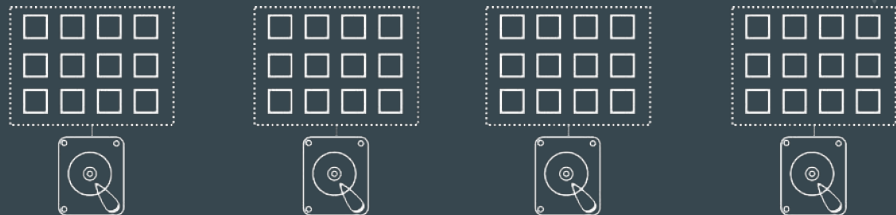
Blockweave Basics: Why are blockweaves better?

- Recall blocks
 - Blockweaves include a recall block (as well as the previous block) with every newly-mined block
 - Recall blocks are pseudo-randomly generated at the time of mining the previous block
 - This incentivises miners to store existing parts of the weave, allowing them to earn rewards by mining new blocks
- Storage distribution
 - Miners incentivised to prioritise blocks stored by few others
 - Doing so lowers their hashing power requirements
 - At a network level, this naturally creates an evenly-spread and self-organising storage of blocks

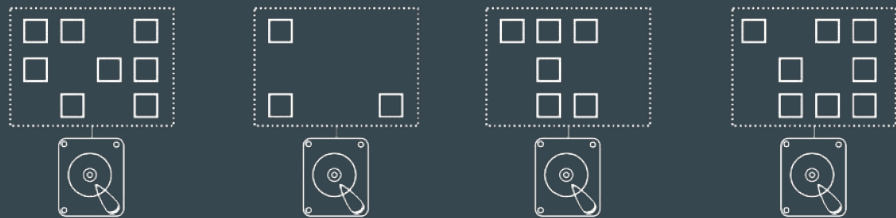


Blockweave Basics: Why are blockweaves better?

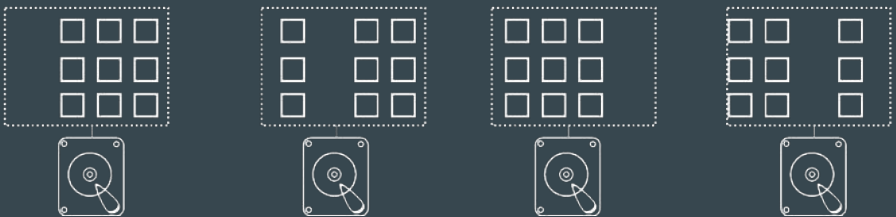
Blockchain



Blockweave
(before self-organisation)



Blockweave
(after self-organisation)





About the Archain project

- A decentralised, permanent archive for the internet
- Built on a shardable blockweave cryptocurrency
- Able to scale to huge sizes
- Cryptographic verification at the points of archiving, storage, and retrieval



Internet archive on the Archain

- Hand-curated record of human history
 - Cryptographically secured on a geographically distributed hard-drive
- Protects the information record
 - Against being lost, damaged, or altered (accidentally or intentionally)
 - Beyond both censorship and natural disaster
 - E.g. Fire at the Internet Archive storage facility
- More than *just* an archive - users will be able to 'step through' versions of the webpages as they existed at different points in time
- Archived pages maintain existing addresses
 - Allowing hyperlinks to be maintained
- Solves the problem of the 'Orwellian memory hole'



Building the Archain project

- Blockweave datastructure
 - Sharding allows for true scaling of the distributed network
- Verification of content
 - Algorithm ensuring clients & miners reach a consensus on webpage content before archiving
- Erlang
 - Automatic distribution of processes
 - Rapid development of prototype network
 - Running large-scale test networks in realistic conditions inside one machine
 - Networking model
 - Message passing and networking treated identically



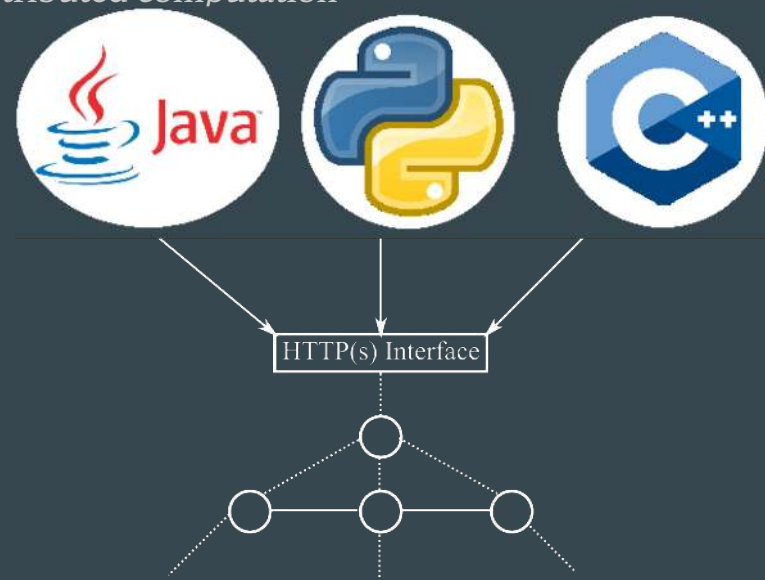
Building the Archain project: Testing the network

- Never-Off Very Large Network Simulator (NO-VLNS)
 - Framework for running massive (1000+ nodes) simulated networks
 - Realistic conditions, latency, packet loss, and network failure
 - Full client simulation
- Tiny Network Tests (TNT)
 - Extreme bug isolation
 - No automatic mining
 - We can create and test highly unusual situations with strong consistency



Apps on the Archain

- Creating a new model for decentralised, permanence-based applications
- Distributed storage
 - Archain apps utilise the blockweave for distributed *storage*
 - Traditional blockchain apps use blockchains for distributed *computation*
- Develop apps in any programming language
 - Via a simple RESTful HTTP or Erlang interface
- Monitor network messages
- Some possible applications:
 - Automated trading
 - Brand management
 - Decentralised social media





The Archain App Development Competition

- Launched on 3rd of November 2017, closes on 3rd August 2018
- Purposes of the competition:
 - Kick-start the Archain app development community
 - Encourage innovative uses of the Archain network & unique blockweave technology
 - Promote the Archain app development ecosystem, by providing developers with small and large volumes of AR to fuel their apps
- Prizes & rewards
 - First prize of 125k AR, two runner up prizes of 62.5k for the best applications submitted
 - Developers are also competing for \$10,000 in investment from the Archain project
 - Plus, all entrants will receive 100 AR for participating!
- Details of requirements & rules can be found on the Archain Medium blog



Questions?

“He who controls the past controls the future, who controls the present controls the past”

- Orwell, 1984.